

5.5

$$a) \text{Var}(b_2|x) = \frac{\sigma^2}{\sum_{t=1}^T (x_t - \bar{x})^2}$$

$$\begin{aligned} \sum_{t=1}^T (t - \bar{t})^2 &= \sum_{t=1}^T t^2 - \frac{(\sum_{t=1}^T t)^2}{T} = \frac{T(T+1)(2T+1)}{6} - \frac{T^2(T+1)^2}{4T} = \frac{T(T+1)}{2} \left[\frac{2T+1}{3} - \frac{(T+1)}{2} \right] \\ &= \frac{T(T^2-1)}{12} \end{aligned}$$

$$\text{Var}(b_2|x) = \frac{\sigma^2}{\frac{T(T^2-1)}{12}} = \frac{12\sigma^2}{T(T^2-1)} \quad \lim_{T \rightarrow \infty} \frac{12\sigma^2}{T(T^2-1)} = 0 \rightarrow \text{consistent}$$

$$b) \sum_{t=1}^T X_t = 0.15 + 0.15^2 + 0.15^3 + \dots + 0.15^T = \frac{0.15(1-0.15^T)}{1-0.15} = 1-0.15^T$$

$$\sum_{t=1}^T X_t^2 = \sum_{t=1}^T (0.15)^t = \frac{1}{3}(1-0.15^T)$$

$$\sum_{t=1}^T (X_t - \bar{x})^2 = \sum_{t=1}^T X_t^2 - \frac{(\sum_{t=1}^T X_t)^2}{T} = \frac{1}{3}(1-0.15^T) - \frac{(1-0.15^T)^2}{T}$$

$$\text{Var}(b_2|x) = \frac{\sigma^2}{\frac{1}{3}(1-0.15^T) - \frac{(1-0.15^T)^2}{T}}$$

$$\lim_{T \rightarrow \infty} \text{Var}(b_2|x) = 3\sigma^2 \rightarrow \text{not consistent}$$

c)

For (a), X increases over time, when the sample size approach infinity, variance approach zero.

For (b), X decays exponentially, variance converges to $3\sigma^2$.

5.6

$$a) H_0: \beta_2 = 0$$

$$H_a: \beta_2 \neq 0$$

$$t = \frac{b_2 - 0}{\text{se}(b_2)} \sim t_{60}$$

$$R.R.: \{t, |t| \geq 2.003\}$$

$$t^* = \frac{3}{\sqrt{4}} = 1.5 \notin R.R. \quad \text{Do not reject } H_0$$

We don't have sufficient evidence to show that β_2 differs from 0.

$$b) H_0: \beta_1 + 2\beta_2 = 5$$

$$H_a: \beta_1 + 2\beta_2 \neq 5$$

$$t = \frac{b_1 + 2b_2 - 5}{\text{se}(b_1 + 2b_2)} \sim t_{60}$$

$$\text{Var}(b_1 + 2b_2)$$

$$= Vb_1 + 4Vb_2 + 4\text{Cov}(b_1, b_2)$$

$$= 3 + 4 \times 4 + 4 \times (-2)$$

$$= 11$$

$$R.R.: \{t, |t| \geq 2.0003\}$$

$$t^* = \frac{8 - 5}{\sqrt{11}} = 0.9 \notin R.R. \quad \text{Do not reject } H_0$$

$$c) H_0: \beta_1 - \beta_2 + \beta_3 = 4$$

$$H_a: \beta_1 - \beta_2 + \beta_3 \neq 4$$

$$t = \frac{(b_1 - b_2 + b_3) - 4}{\text{se}(b_1 - b_2 + b_3)} \sim t_{60}$$

$$\text{Var}(b_1 - b_2 + b_3)$$

$$= Vb_1 + Vb_2 + Vb_3 - 2\text{Cov}(b_1, b_2) - 2\text{Cov}(b_1, b_3) + 2\text{Cov}(b_2, b_3)$$

$$= 16$$

$$R.R.: \{t, |t| \geq 2.003\}$$

$$t^* = \frac{(2 - 3 - 1) - 4}{\sqrt{16}} = \frac{-6}{4} = -1.5 \notin R.R.$$

Do not reject H_0 .

b2o

$$a. \bar{y}_j = \beta_1 + \beta_2 \bar{x}_j + \bar{e}_j$$

$$\bar{y}_2 = \beta_1 + \beta_2 \bar{x}_2 + \bar{e}_2$$

$$\hat{\beta}_{2, \text{mean}} = \frac{(\beta_1 + \beta_2 \bar{x}_2 + \bar{e}_2) - (\beta_1 + \beta_2 \bar{x}_1 + \bar{e}_1)}{\bar{x}_2 - \bar{x}_1} = \beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}$$

$$E(\hat{\beta}_{2, \text{mean}} | X) = \beta_2, \quad E(\bar{e}_1 | X) = 0$$

$\hat{\beta}_{2, \text{mean}}$ is unbiased estimator of β_2

$$b. \text{Var}(\hat{\beta}_{2, \text{mean}} | X) = \text{Var}\left(\beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}\right)$$

$$= \frac{\frac{2\sigma^2}{N} + \frac{2\sigma^2}{N}}{(\bar{x}_2 - \bar{x}_1)^2}$$

$$= \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}$$

$$c. \text{Var}(\hat{\beta}_{2, \text{mean}}) = E[\text{Var}(\hat{\beta}_{2, \text{mean}} | X)] + V[E(\hat{\beta}_{2, \text{mean}} | X)] = E\left[\frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}\right]$$

$$d. E(\hat{\beta}_{2, \text{mean}} | X) = \beta_2 \rightarrow \text{consistent}$$

$$\lim_{N \rightarrow \infty} V(\hat{\beta}_{2, \text{mean}}) = 0$$

When $N \rightarrow \infty$, we find $X \rightarrow N(\bar{x}_2 - \bar{x}_1)^2 \rightarrow \infty$, then $\hat{\beta}_{2, \text{mean}}$ is consistent estimator.